



Dynamical Analysis of Cyber Attacks on Power System Cascading Failures

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EE4410 - Cyber Security of Power Grids

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Outline

1. Time-Frequency Analysis

- Motivation
- Hilbert-Huang Transform (HHT)
- Empirical Mode Decomposition (EMD)

2. Point of no return identification and quantification

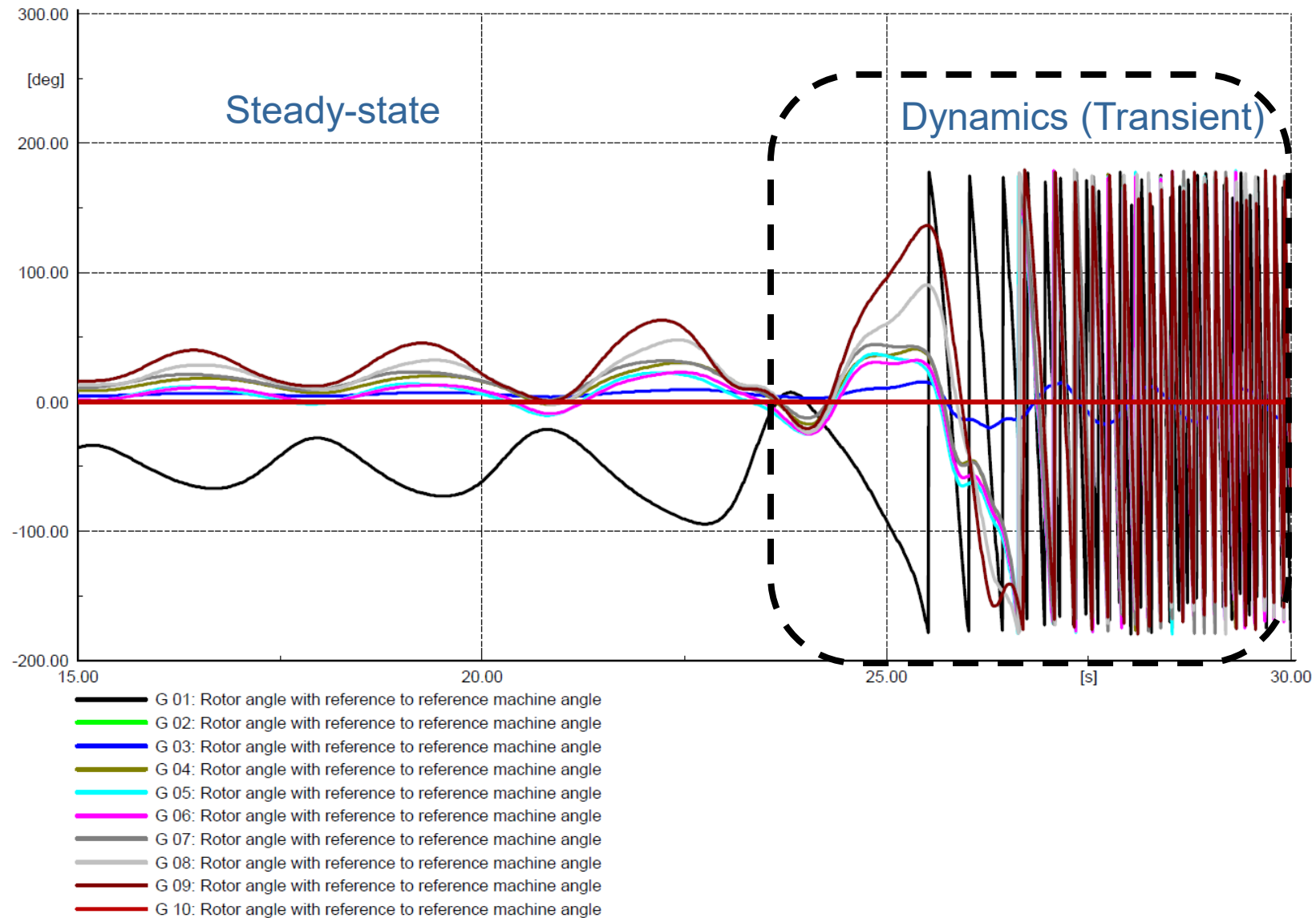
- Application of HHT for cascading failure analysis

3. Experimental results



1. Time-Frequency Analysis

Power System Dynamics



Numerical Measurement Based Techniques

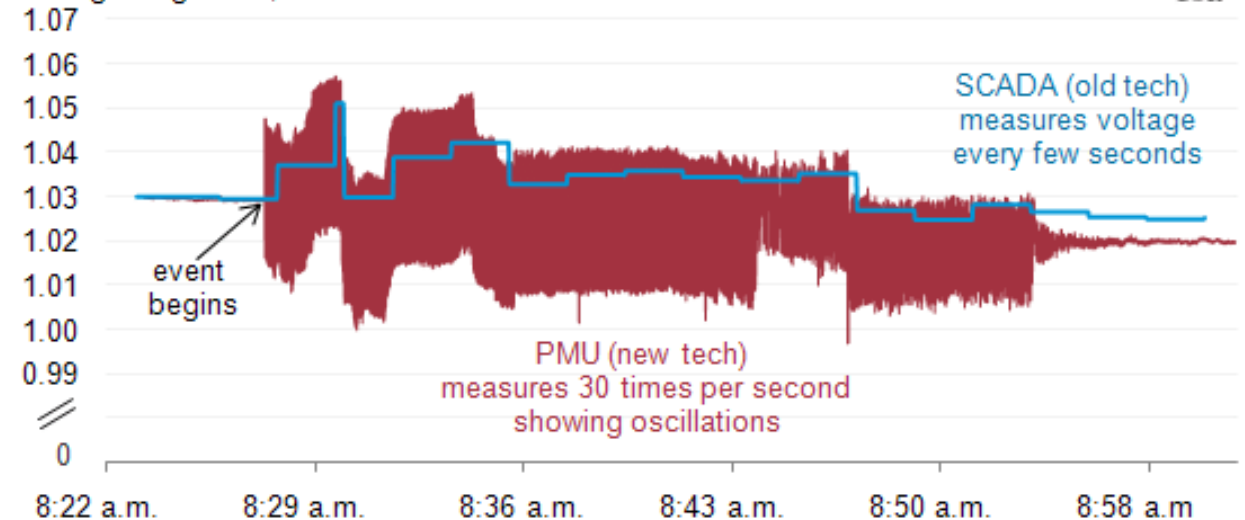
- Why? – analytical solutions not always feasible
- Number of state variables increases with system size
- Numerical solvers/tools provide wealth of data



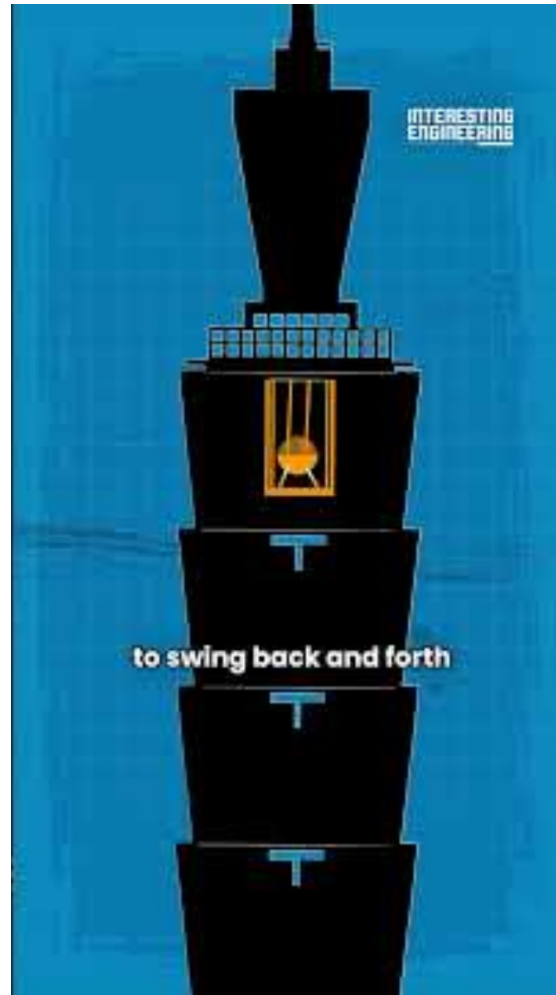
VS



PMU data reveal dynamic behavior as the system responds to a disturbance
Data comparison example, voltage disturbance on April 5, 2011
voltage magnitude, indexed

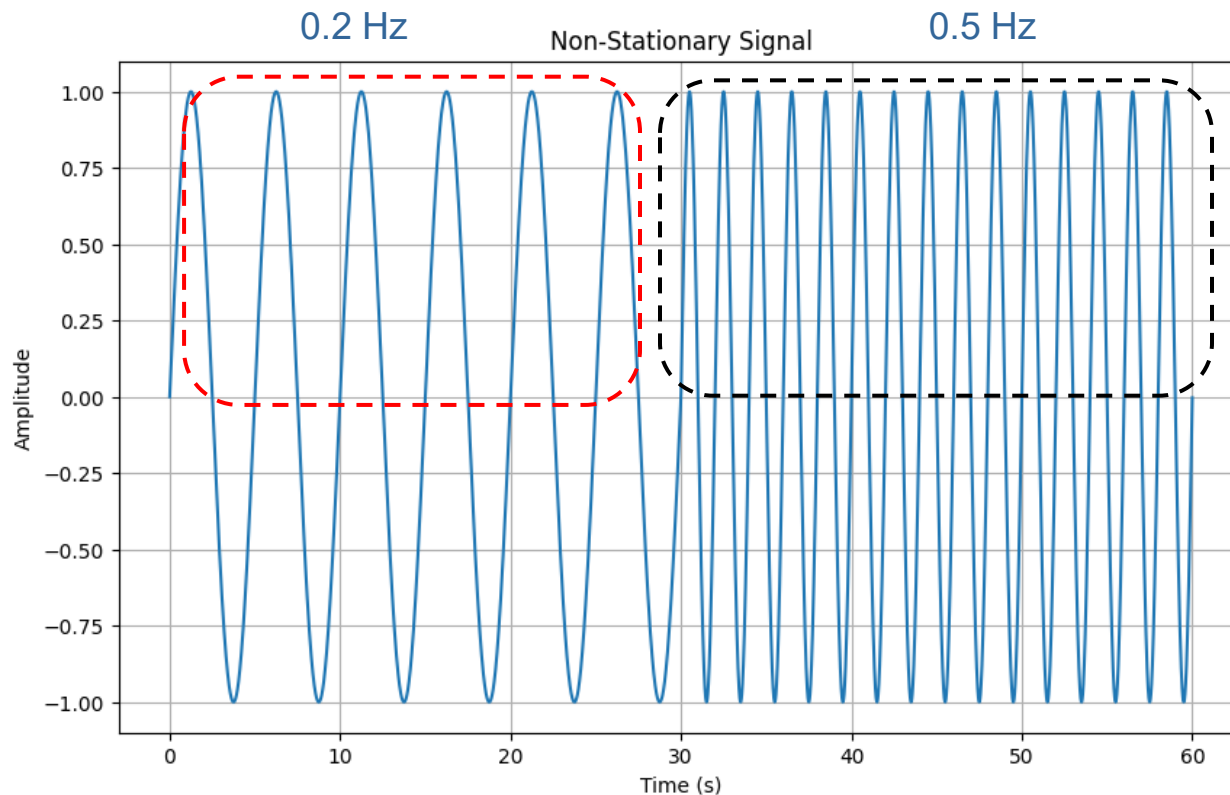


Why Frequency Analysis?

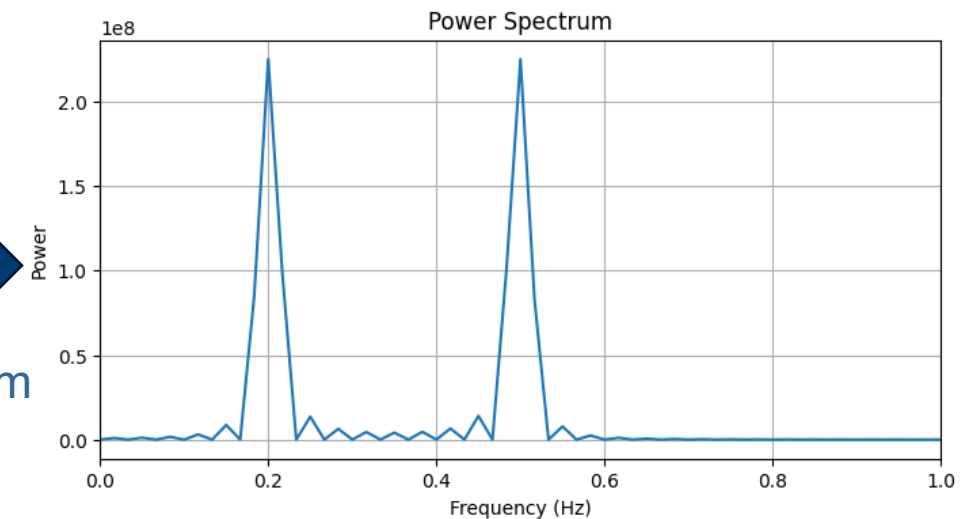


Limitations of Frequency Analysis

- Fourier transform/analysis powerful but has limitations for nonstationary phenomena
- Time information lost, i.e., when do the frequency peaks occur?

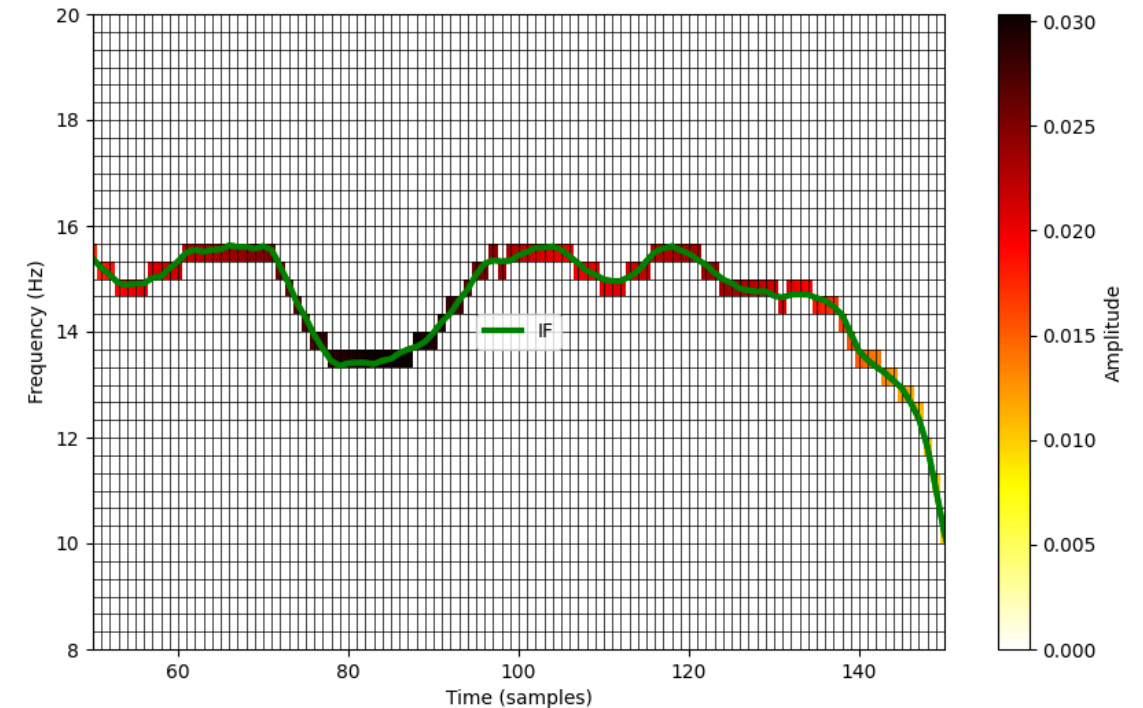


Fourier
transform



Time-Frequency Analysis

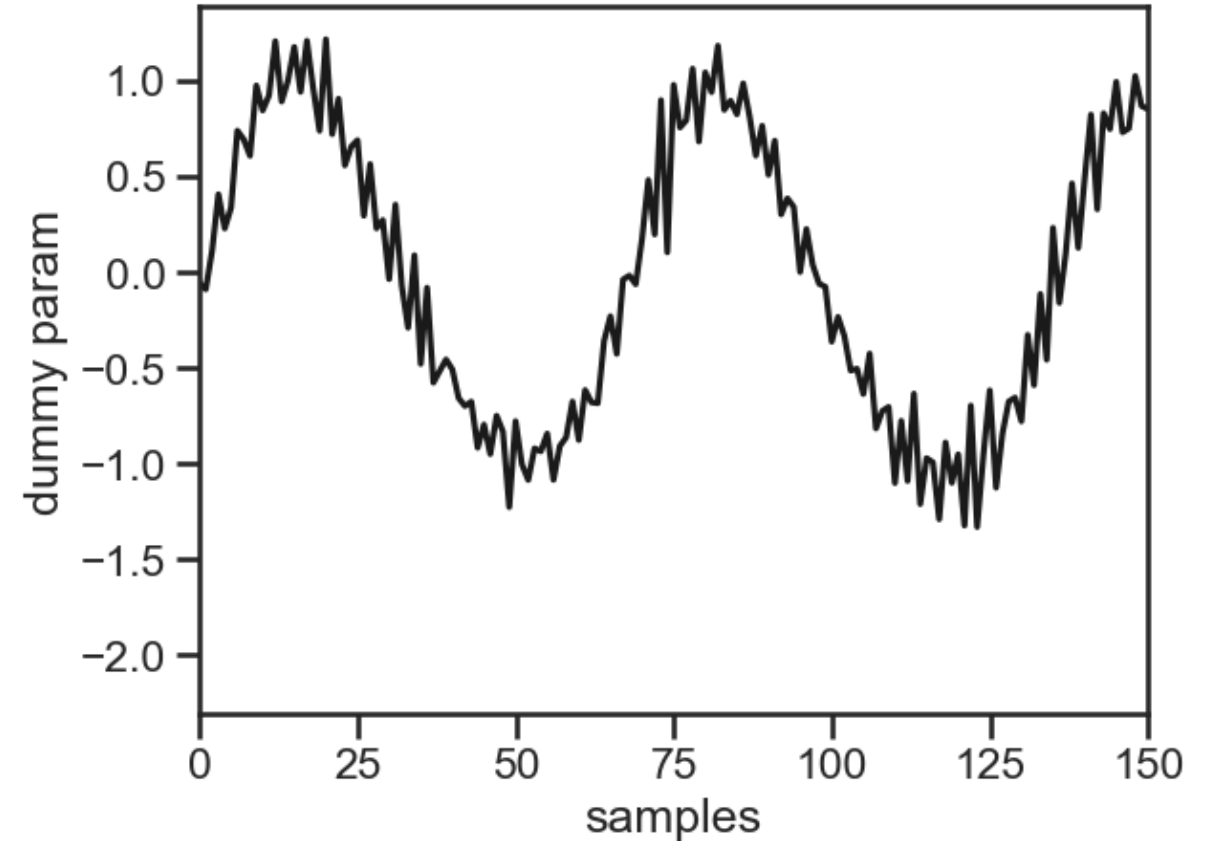
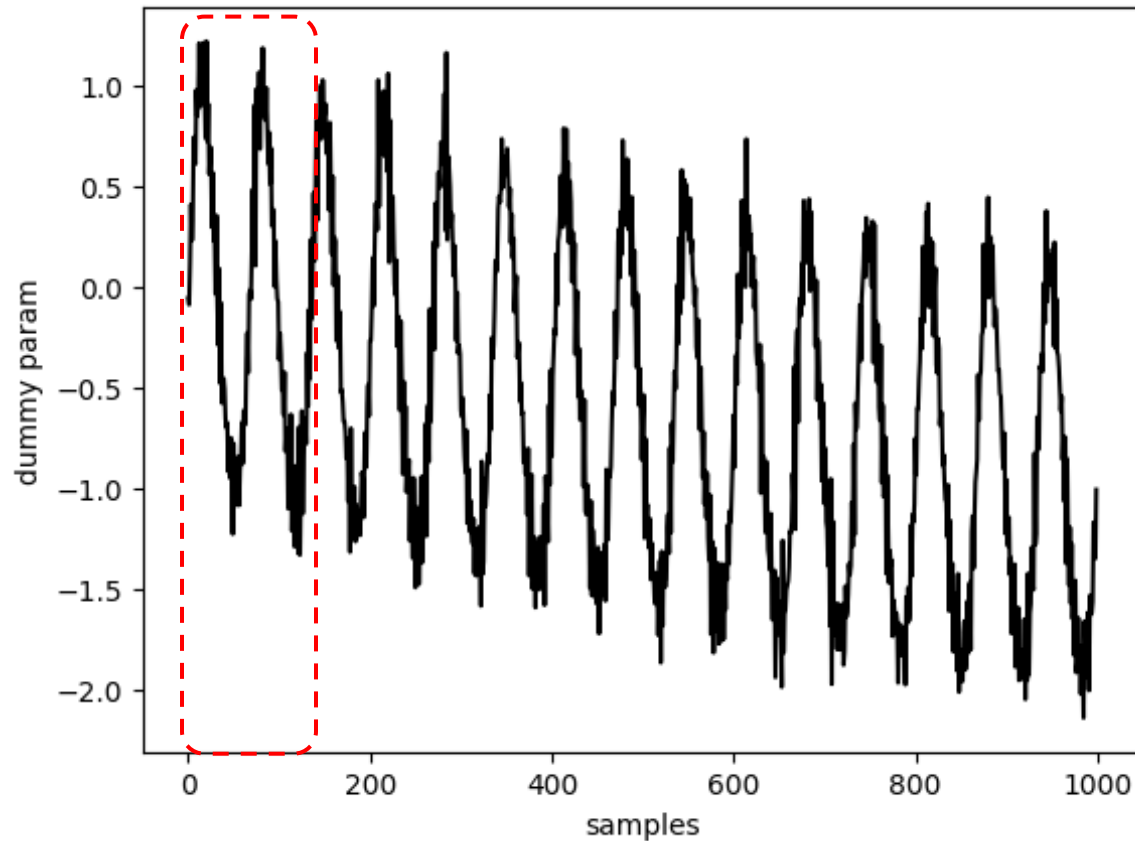
- Capture frequency as well as time-varying behaviour of original signal (absent in Fourier transforms)
- Popular method is Hilbert-Huang Transform
- Two-part strategy:
 1. Empirical Mode Decomposition (EMD)
 2. Hilbert Transform
- Data-driven, empirical method
 - No fundamental theoretical foundations like Fourier



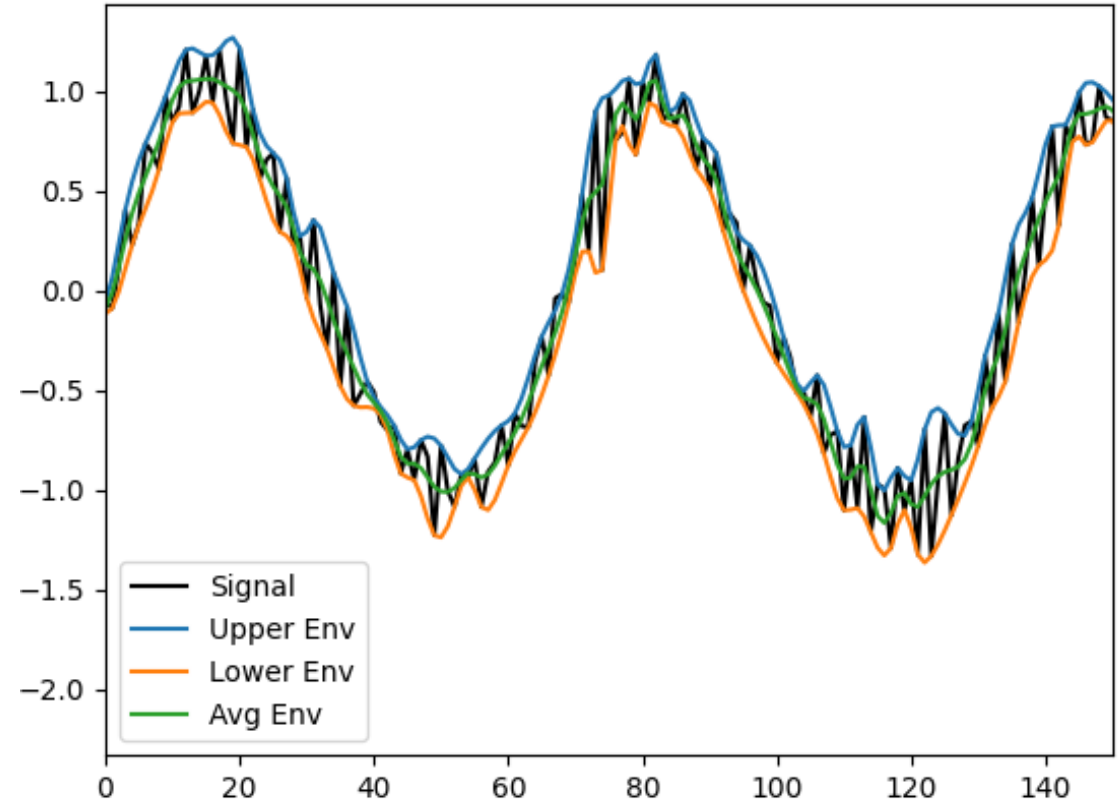
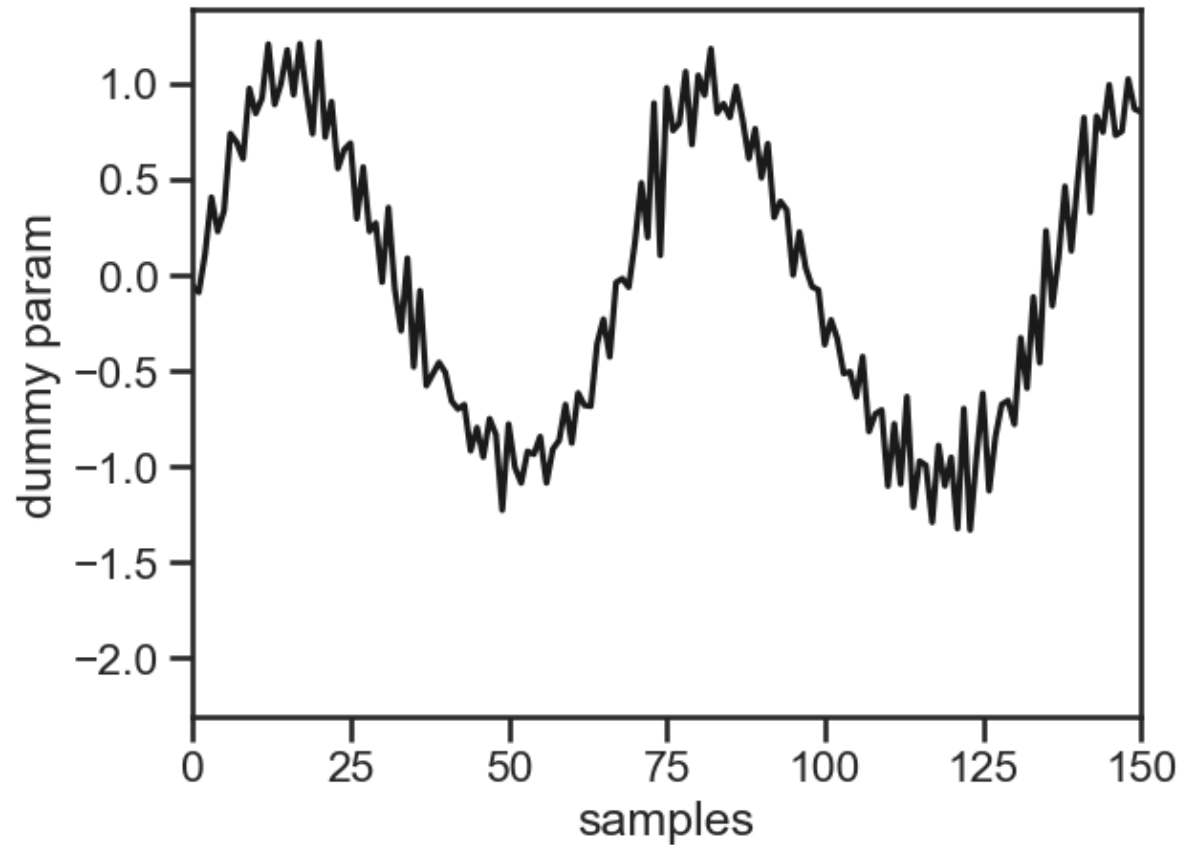
Empirical Mode Decomposition: Sifting

- Iteratively extract oscillatory components from an input signal
- Starting from the fastest and through to the very slowest until only a non-oscillatory trend is left
- Each component is then known as an 'Intrinsic Mode Function' or IMF
 1. Find all local extrema in the signal
 2. Join all the local maxima with a cubic spline, creating an upper envelope. Repeat for local minima and create a lower envelope
 3. Calculate the mean of the envelopes
 4. Subtract mean from original signal
 5. Repeat steps 1-4 until result is an IMF - avg upper and lower envelopes ≈ 0
 6. Subtract this IMF from the original signal
 7. Repeat steps 1-6 till there are no more IMFs left in the signal

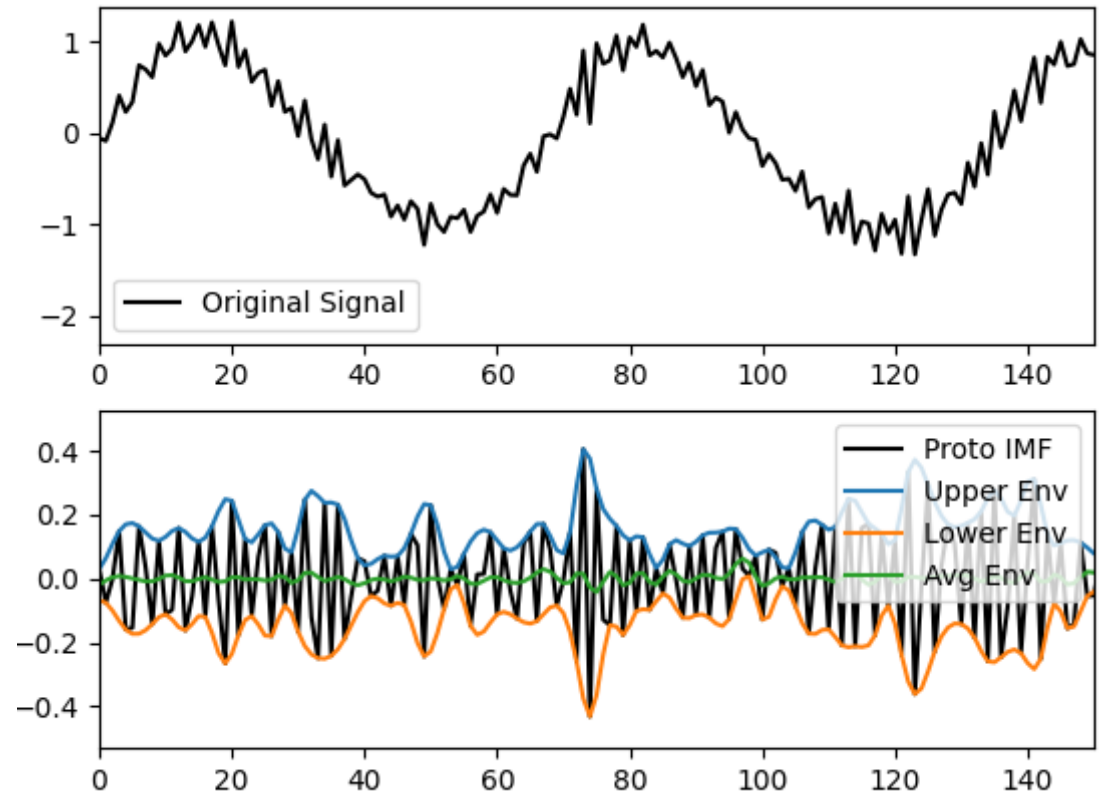
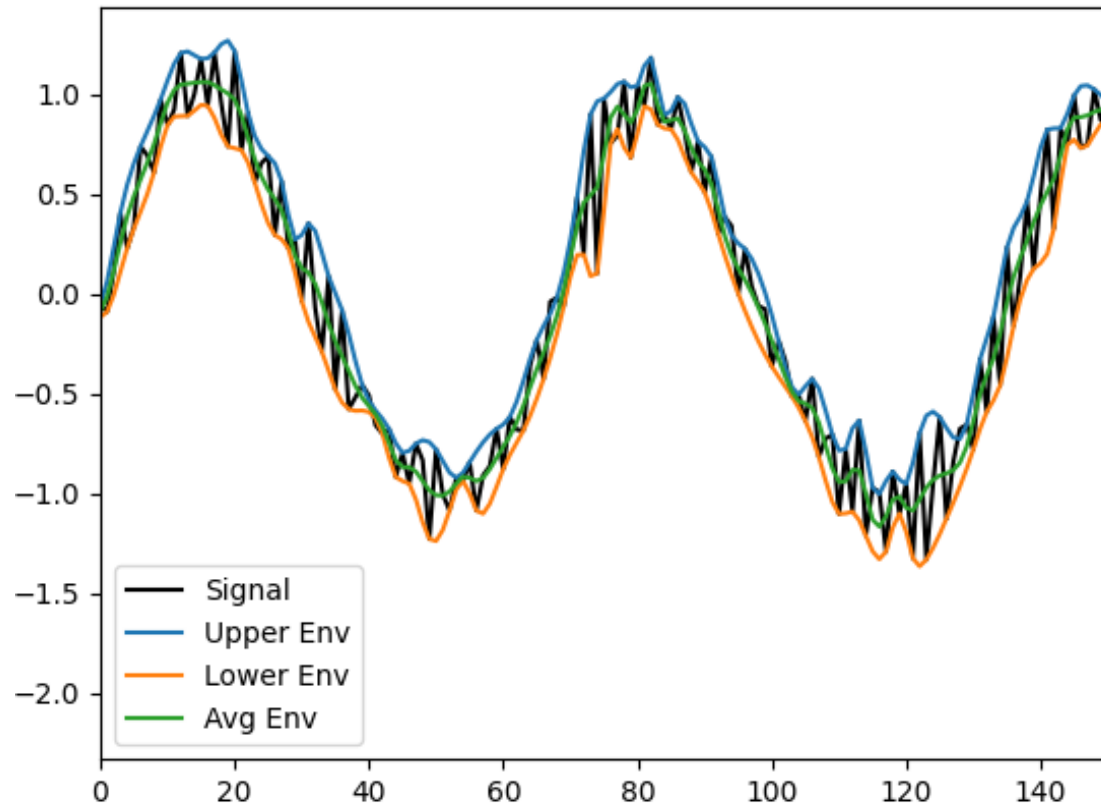
Sifting Example



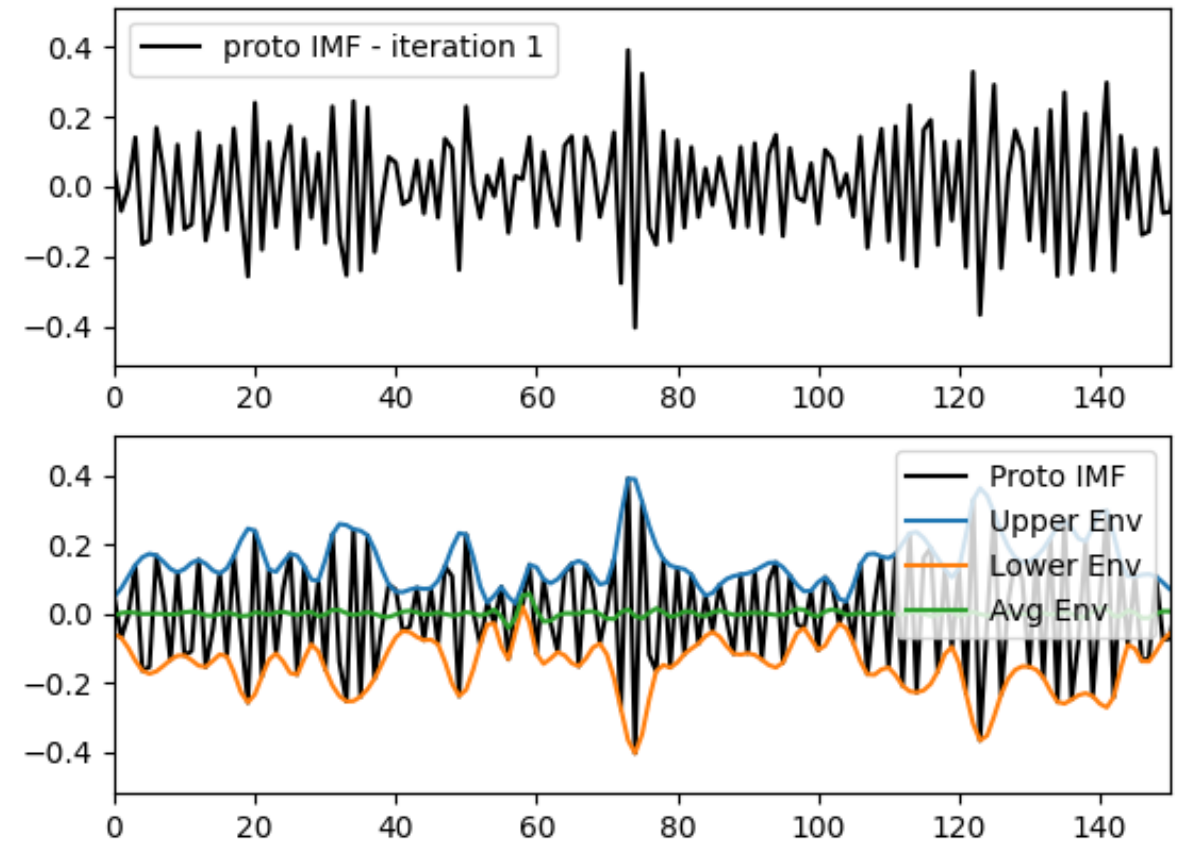
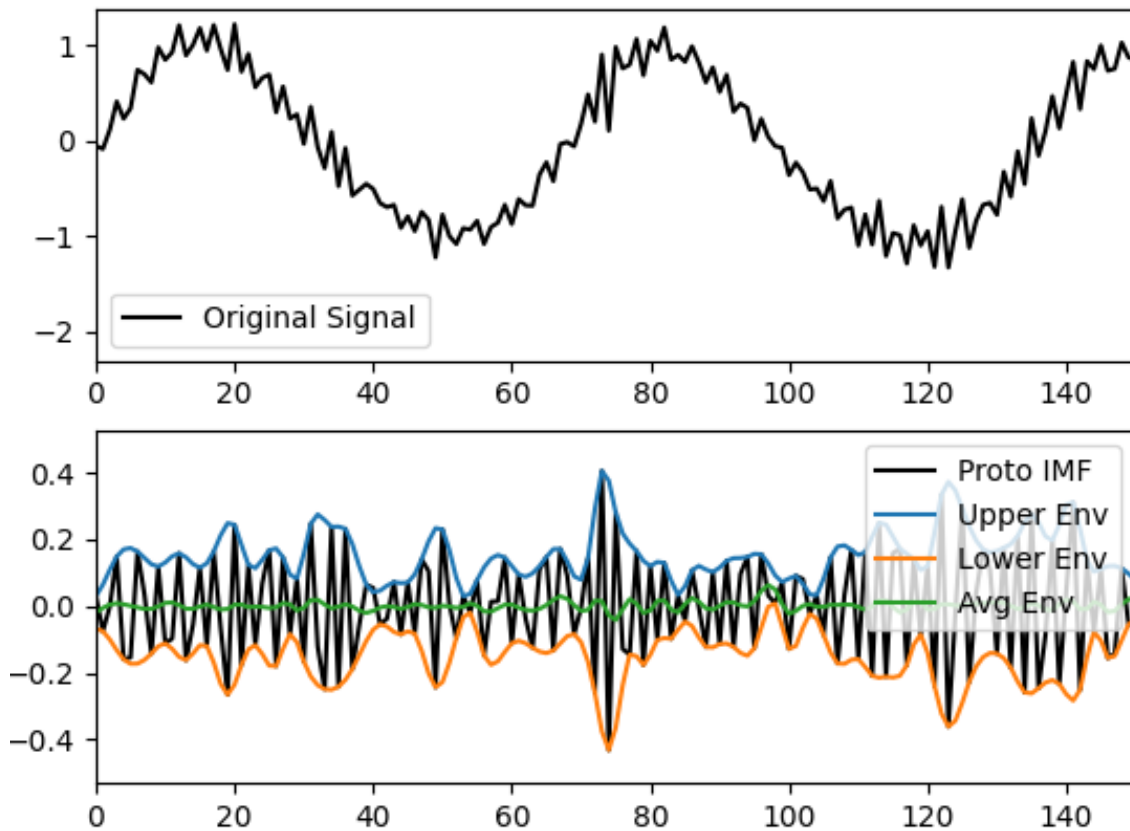
Sifting Example: Envelopes



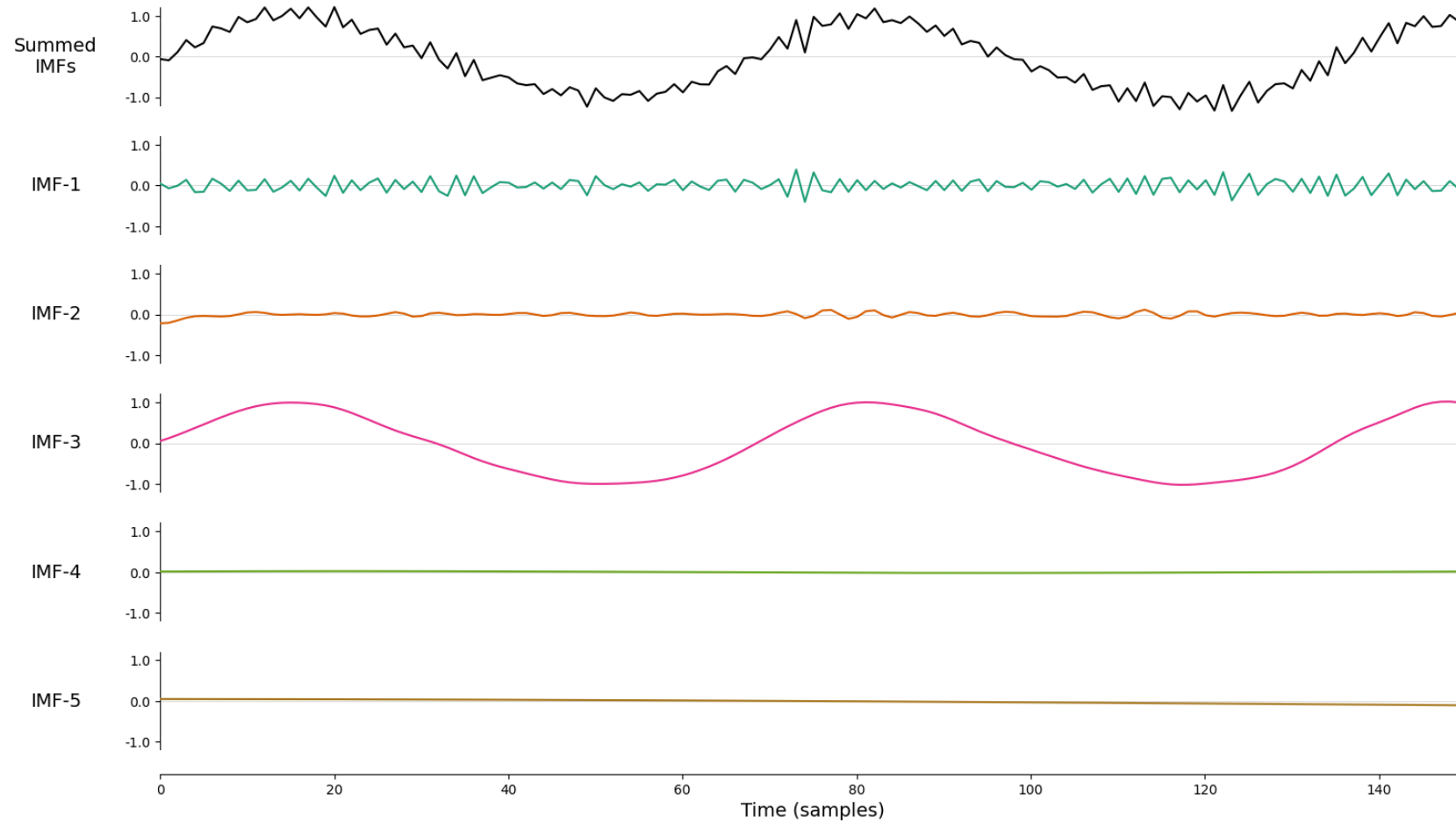
Sifting Calculation



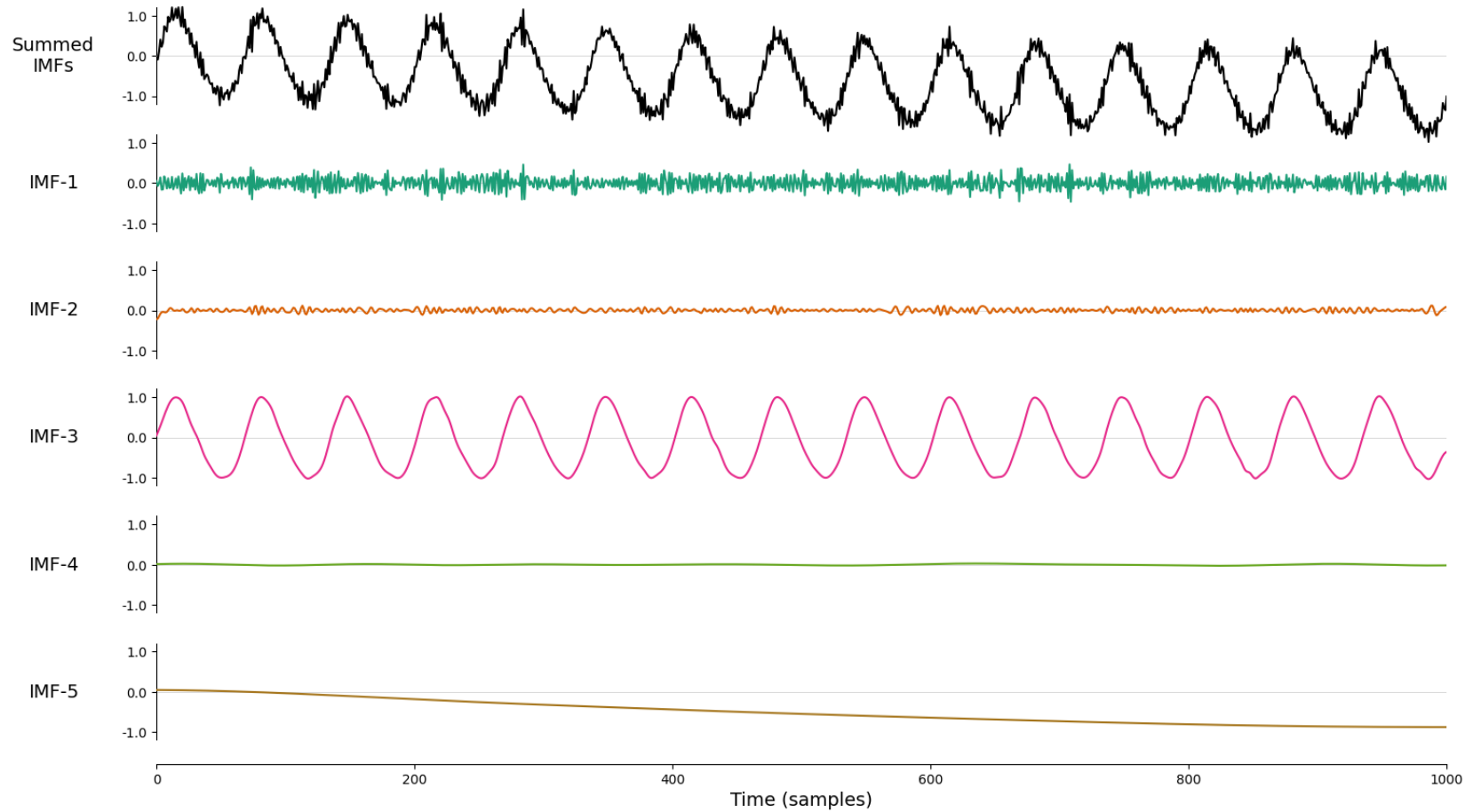
Sifting Calculation



Sifting Result: Zoomed



Sifting Result



Hilbert Transform

- Creates an '*analytic*' form of a signal that is useful for extracting dynamic information from it
- For any real-valued signal $x(t)$, Hilbert transform, $\widehat{x(t)}$ is given by

$$\widehat{x(t)} = x(t) \times \frac{1}{\pi t}$$

- Physical meaning: **shifting phase angles of frequency components**
 - Negative frequency component: $+\frac{\pi}{2}$ (90°)
 - Positive frequency component: $-\frac{\pi}{2}$ (90°)

Analytical Signal

- For any real-valued signal $x(t)$, analytical signal (complex-valued) is defined as :

$$x_a(t) = x(t) + j\widehat{x(t)}$$

Where, $\widehat{x(t)}$ is Hilbert transform

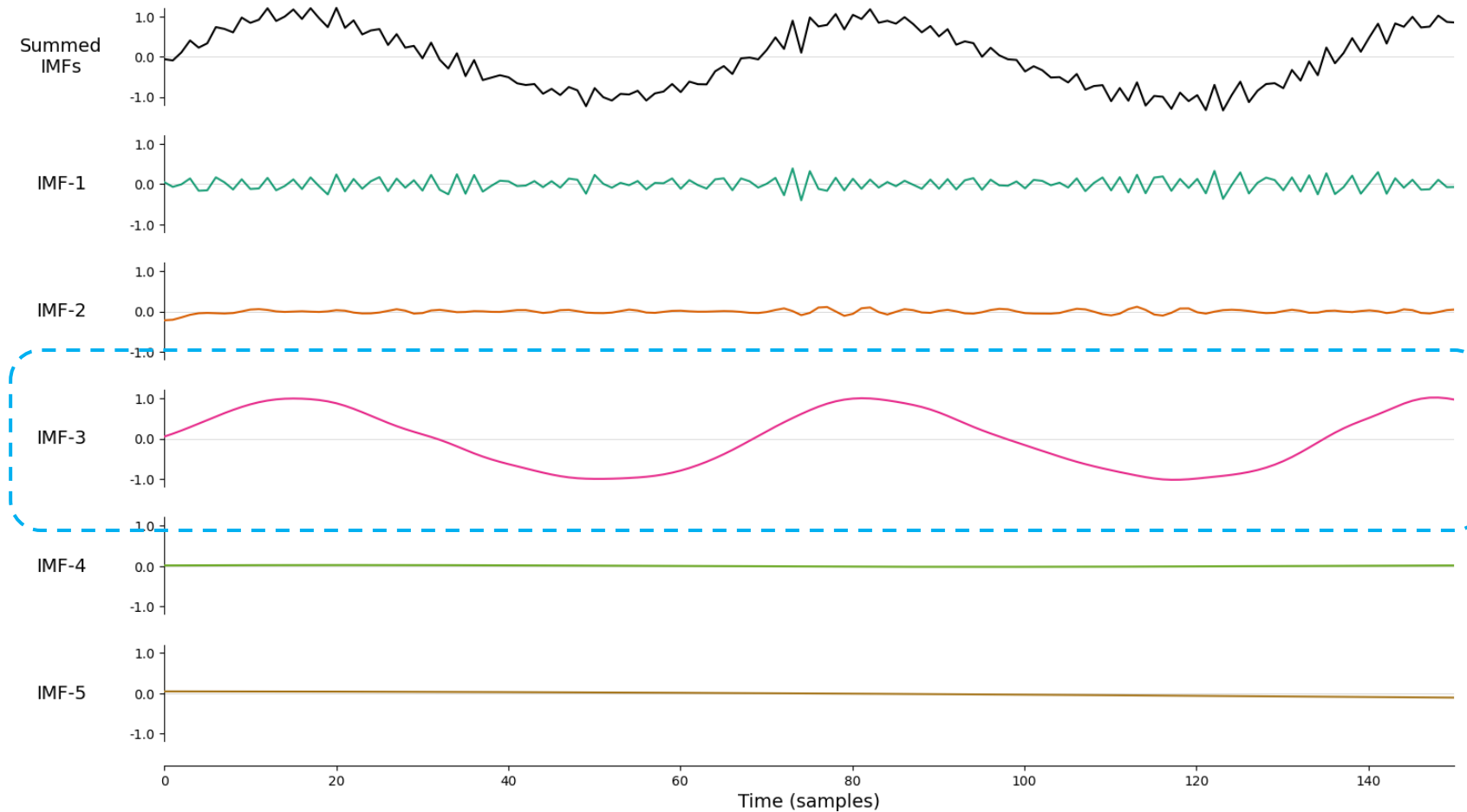
- Instantaneous frequency

$$f_{inst} = \frac{1}{2\pi} \frac{d}{dt} \tan^{-1}[x_a(t)], \text{ aka, rate of change of phase}$$

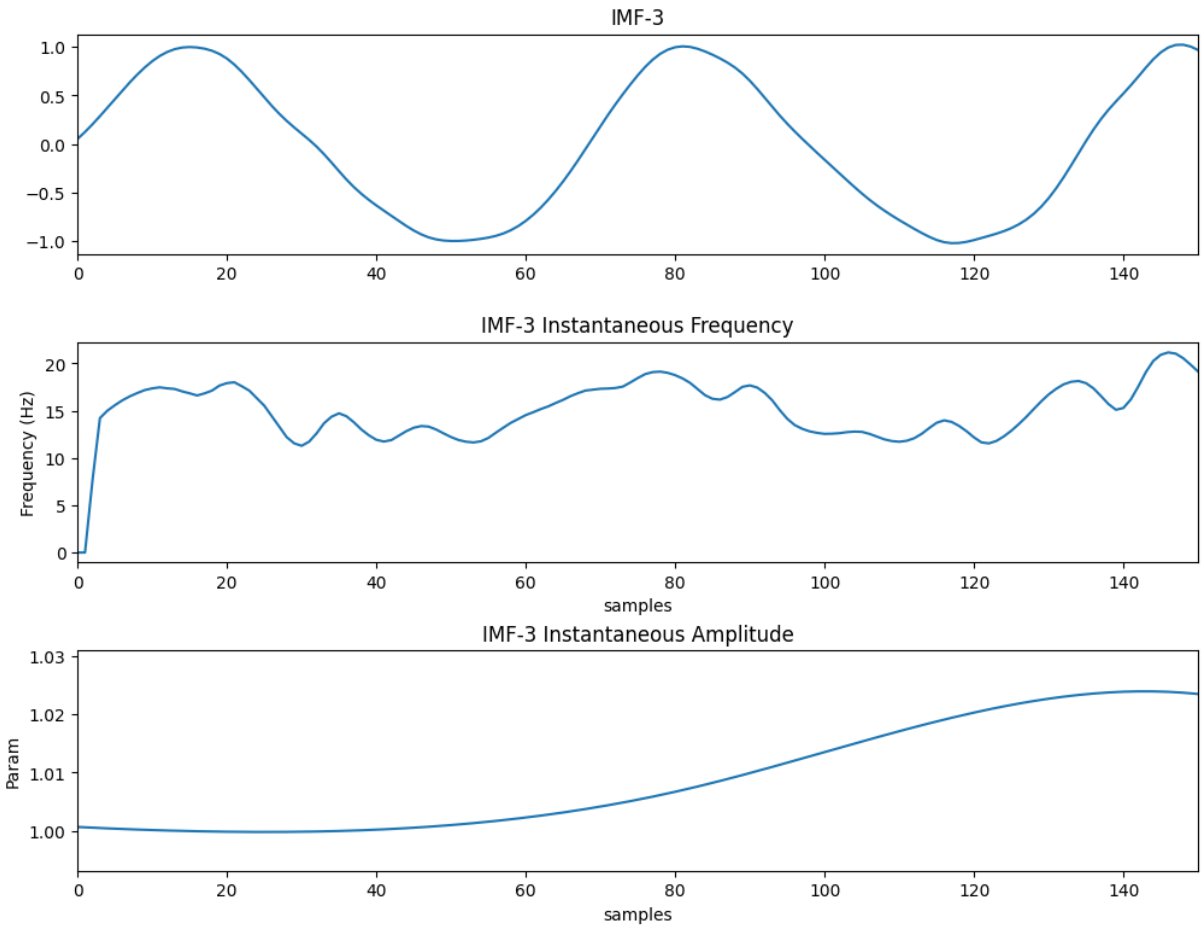
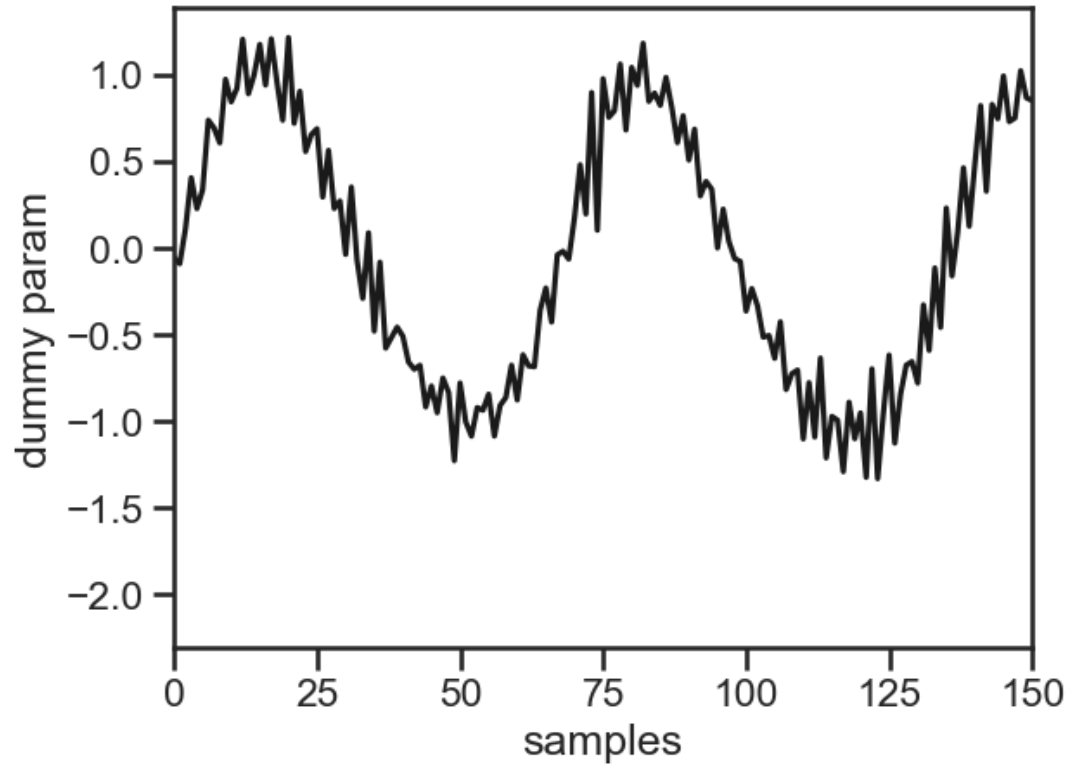
- Instantaneous amplitude

$$A(t) = |x_{a(t)}| = \sqrt{x(t)^2 + \hat{x}(t)^2}$$

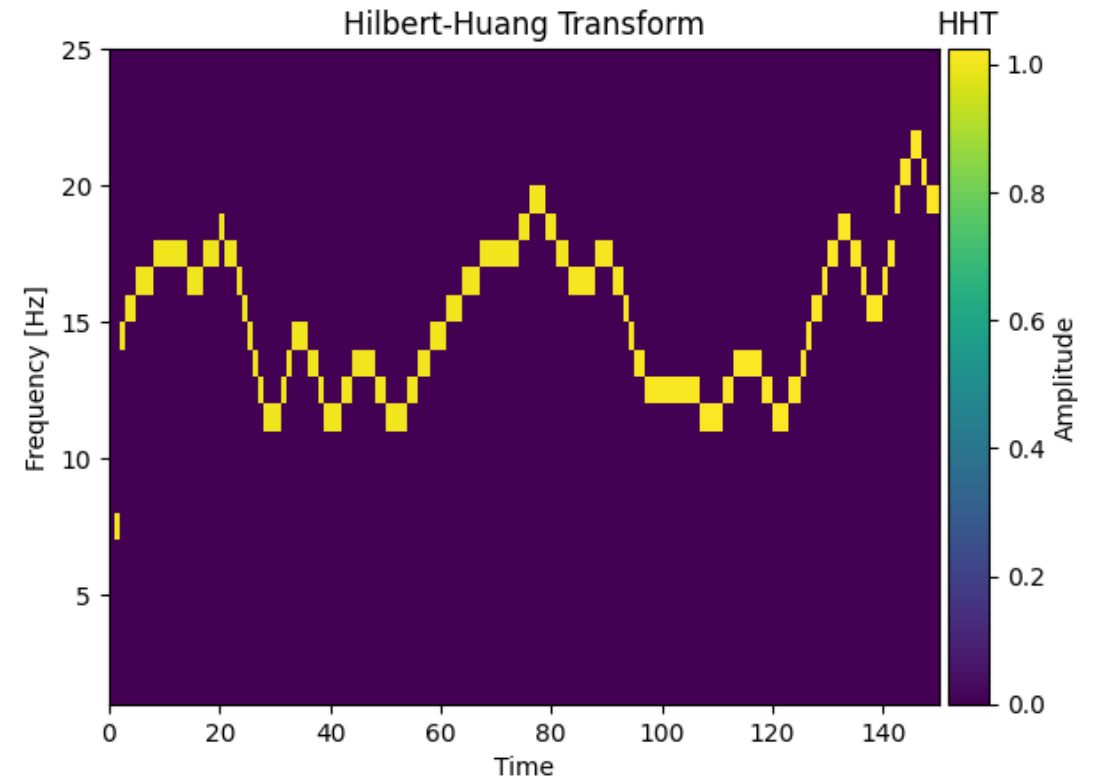
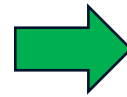
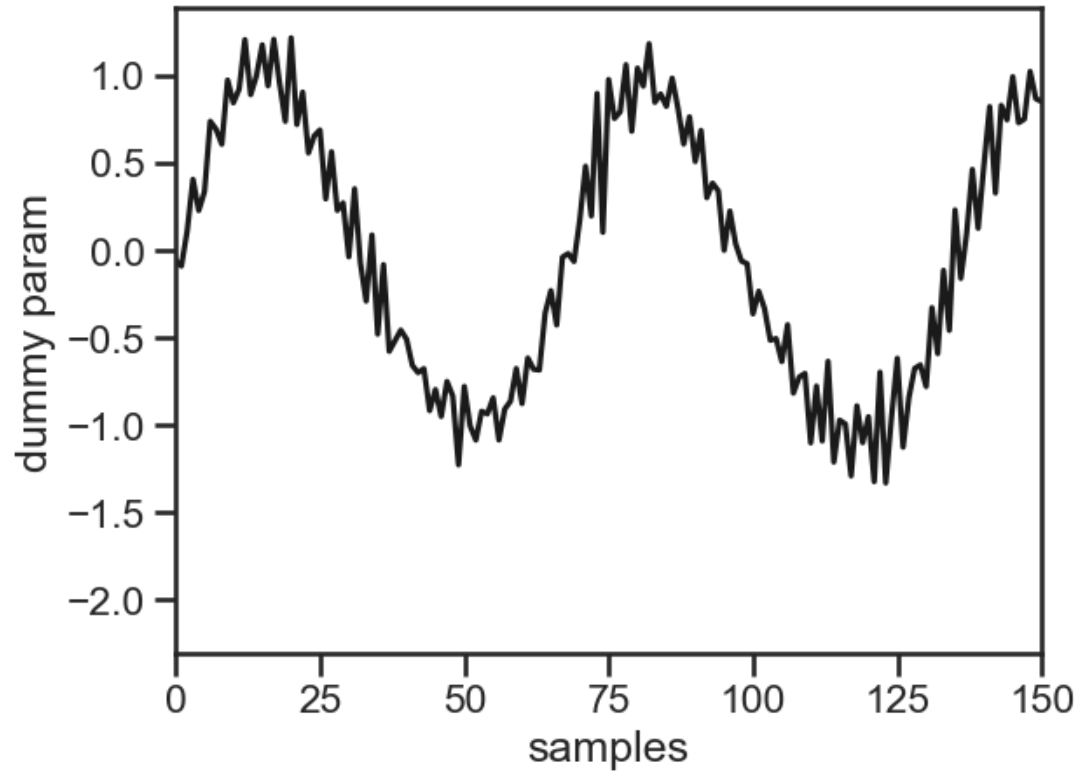
Sifting Result



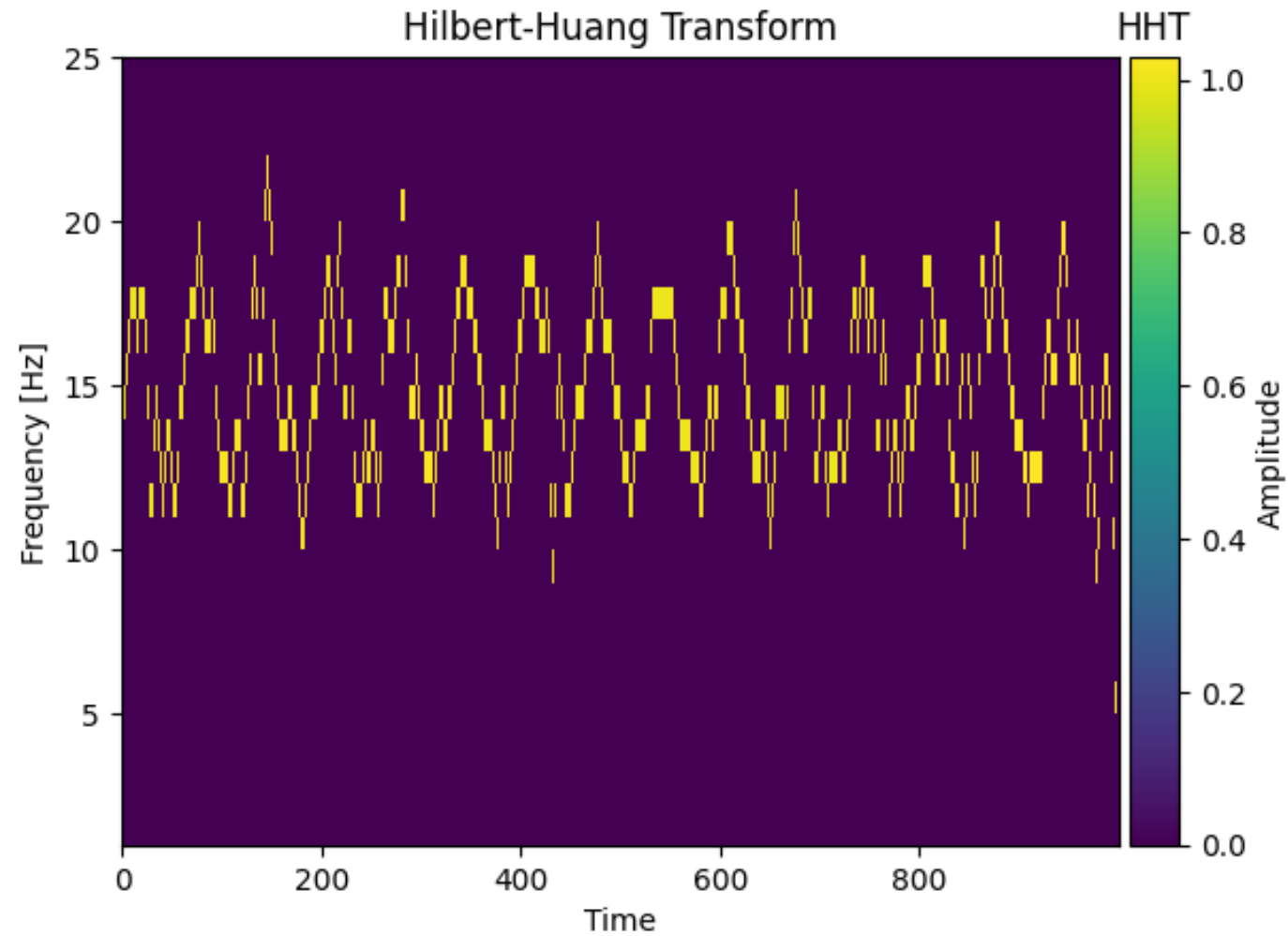
Instantaneous Frequency & Amplitude



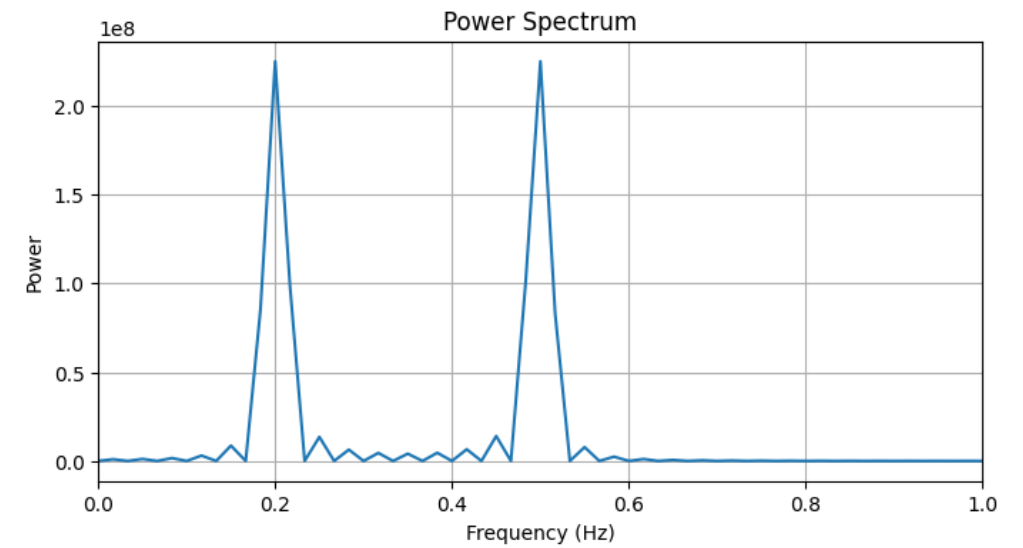
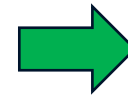
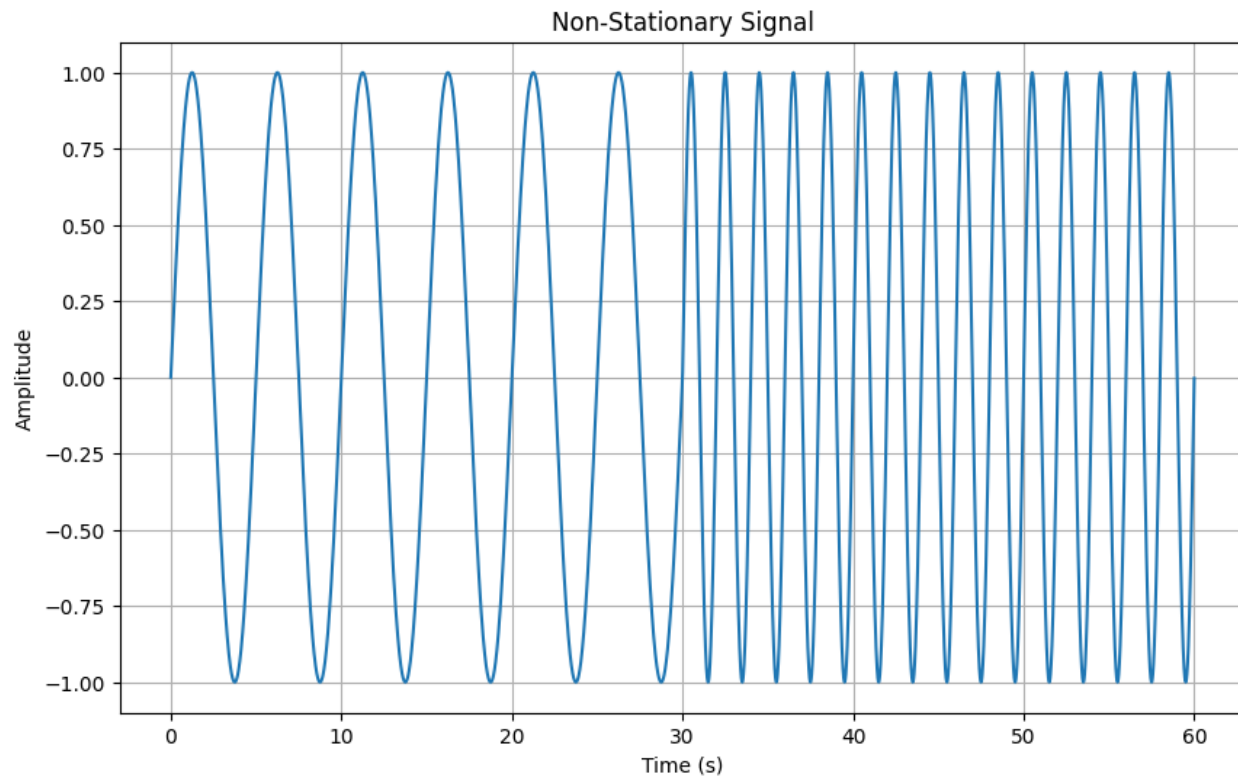
Result of Hilbert-Huang Transform for IMF 3



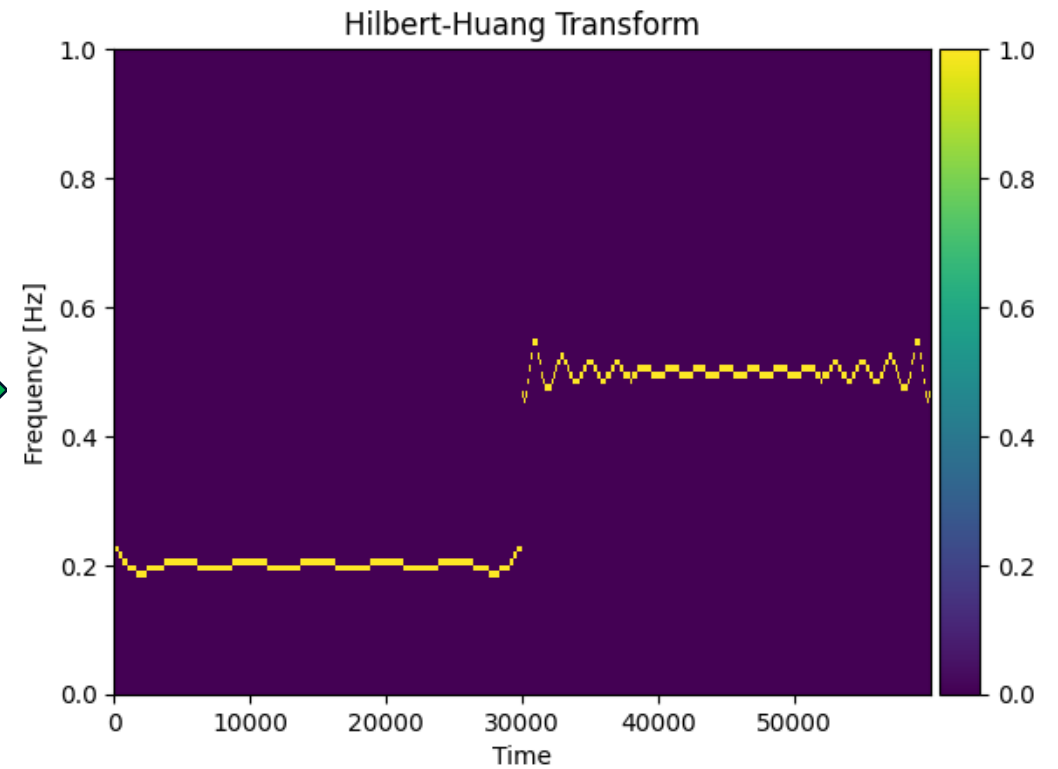
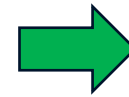
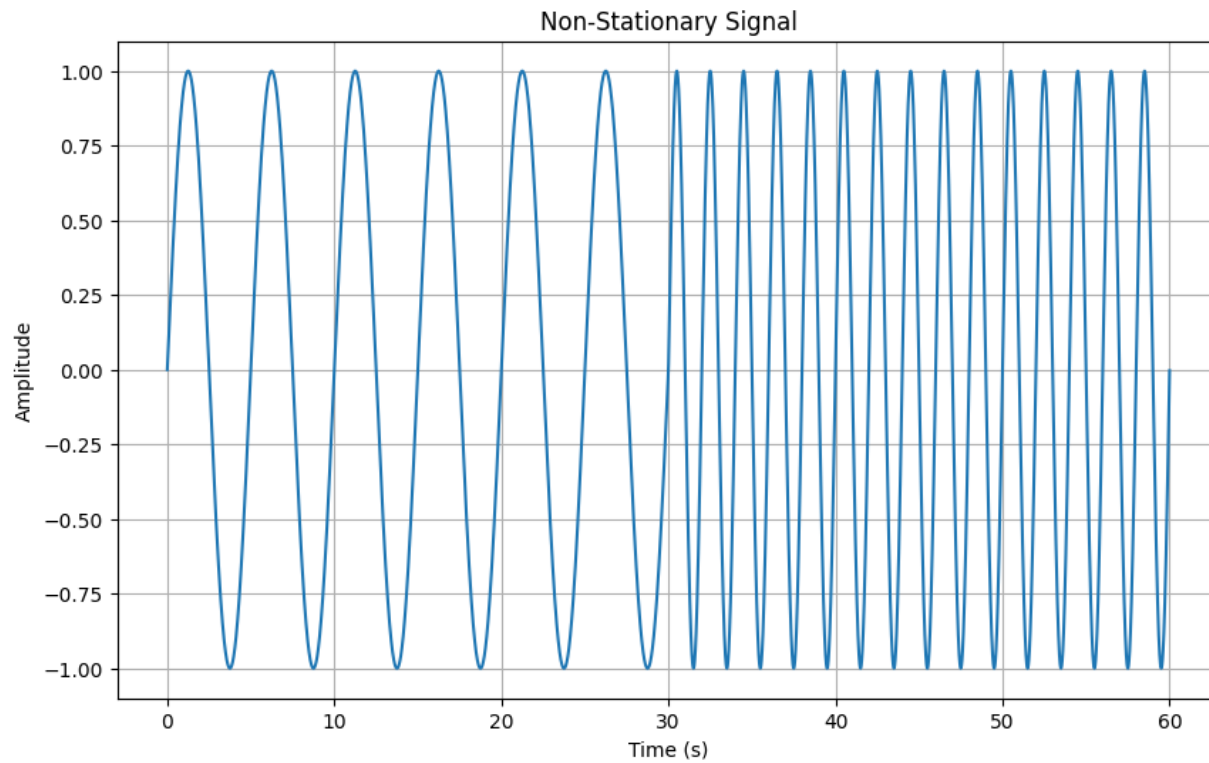
HHT Result: IMF 3



Comparison of FFT vs HHT



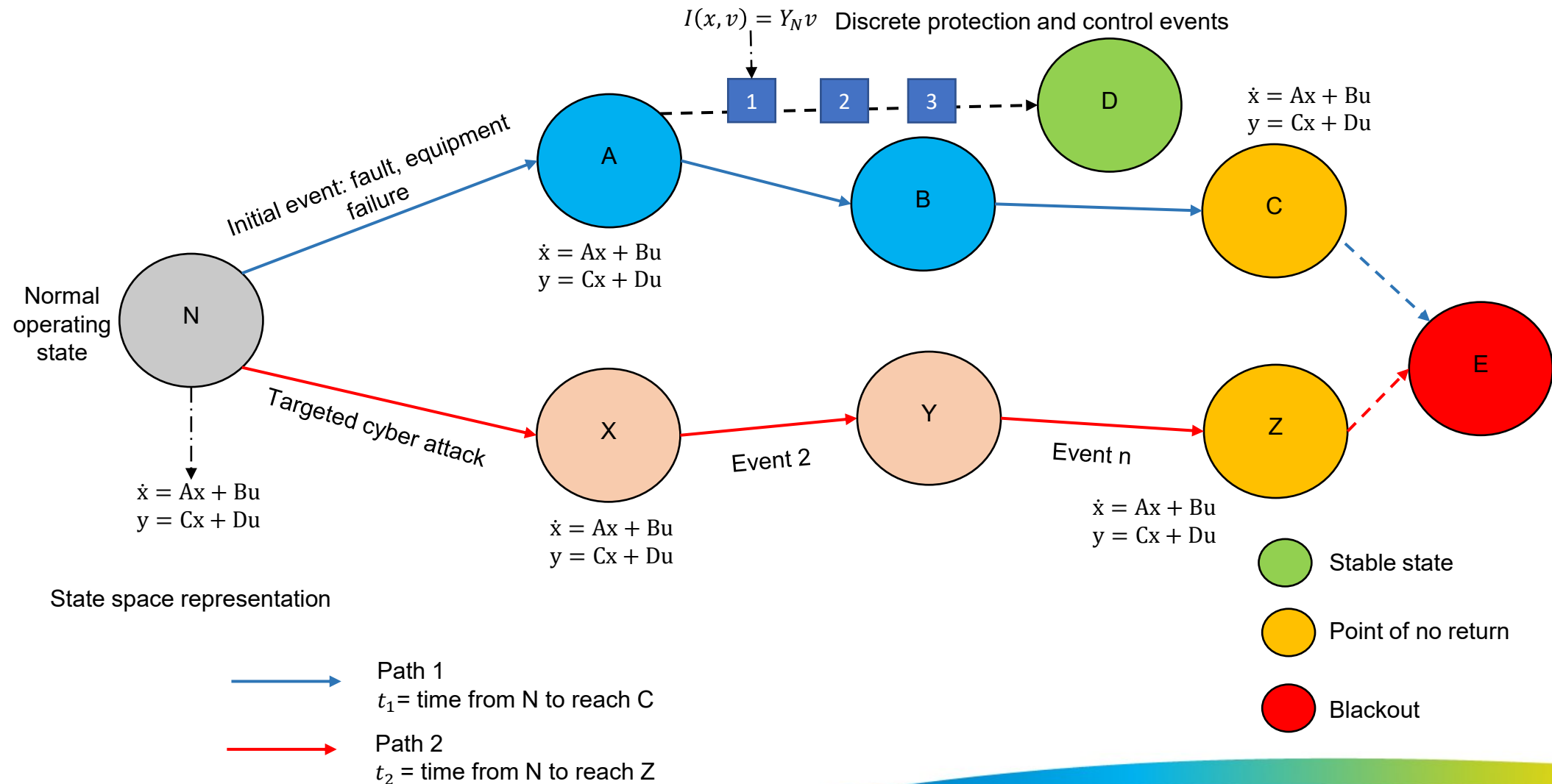
Comparison of FFT vs HHT



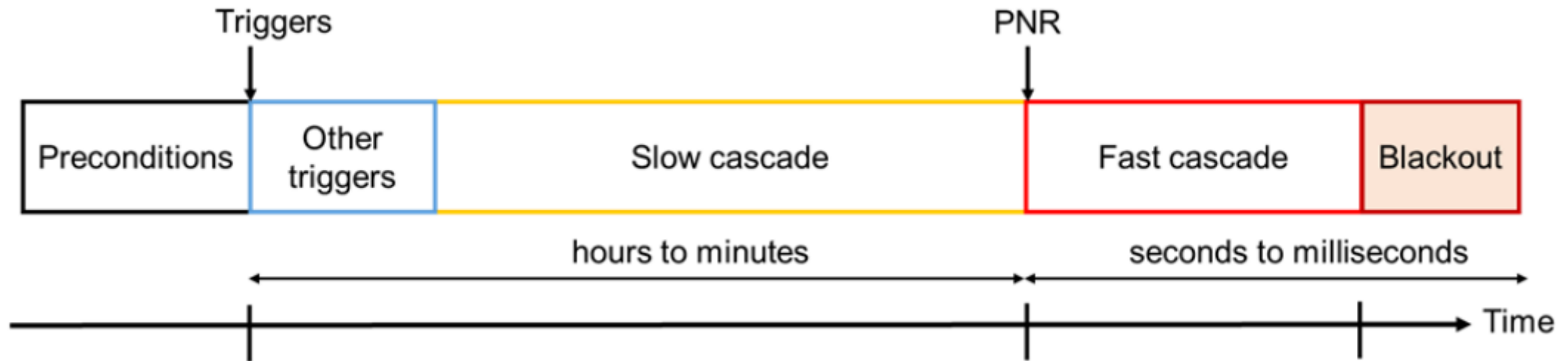


2. Point of No Return

Cascading Failures – State Trajectory

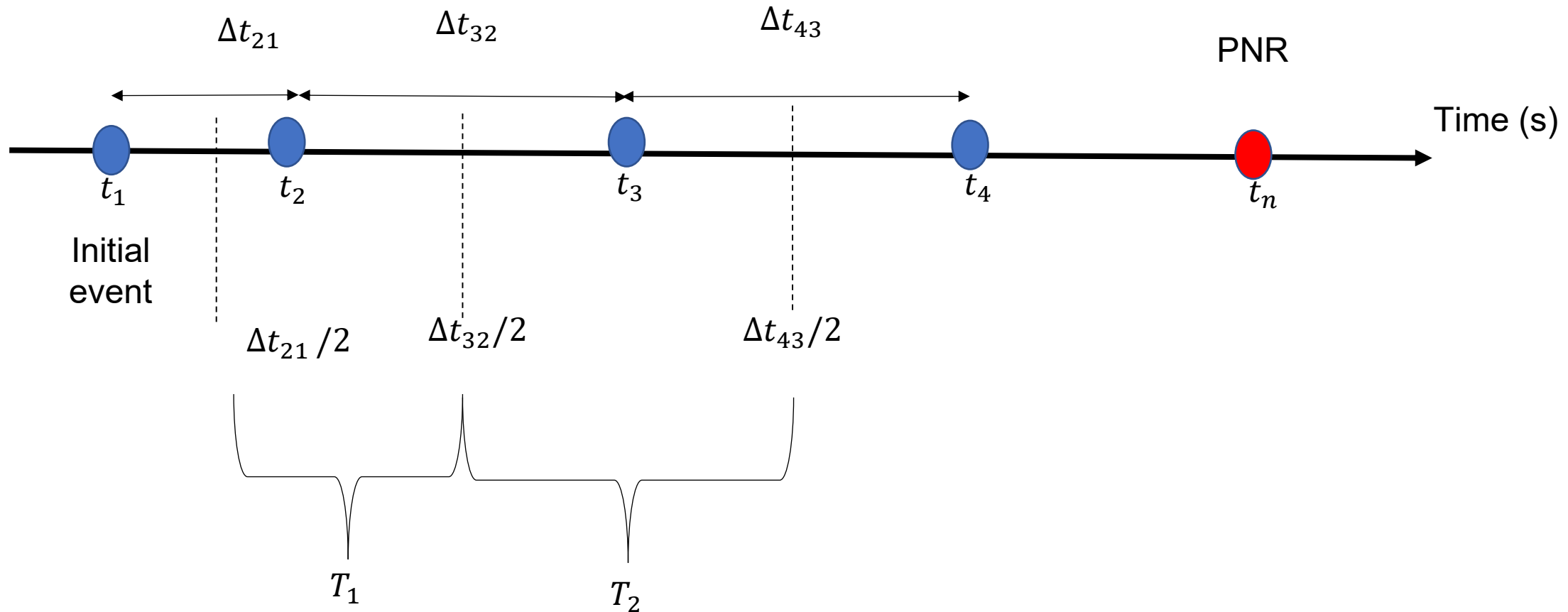


Point of No Return (PNR)



- Historical blackouts occurred usually when the system was already stressed, due to maintenance, etc.
- Slow cascade phase can last for many minutes to hours
- Fast cascade phase lasts a few minutes

Point of No Return (PNR)



Instantaneous Damping Correlation

- Inspiration: similarities between mass-spring and swing equation

$$m \frac{d^2 x}{dt^2} + c \frac{dx}{dt} + kx = 0$$

$$\frac{d(\Delta\bar{\omega}_r)}{dt} - \frac{1}{2H} (\bar{T}_m - \bar{T}_e - K_D \Delta\bar{\omega}_r) = 0$$

- Instantaneous damping:
$$\alpha(t) = -\frac{\dot{A}(t)}{A(t)}$$
- Goal: estimate **point of no return** during **cascading failures** by looking at variations in instantaneous damping

Methodology: Covariance Matrix

- Form a covariance matrix ($n \times n$) for each time window of the instantaneous damping
- Shows how a group of n parameters vary w.r.t each other

$$R = \begin{bmatrix} \sigma(\alpha_1, \alpha_1) & \sigma(\alpha_1, \alpha_2) & \dots & \sigma(\alpha_1, \alpha_n) \\ \sigma(\alpha_2, \alpha_1) & \sigma(\alpha_2, \alpha_2) & \dots & \sigma(\alpha_2, \alpha_n) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma(\alpha_n, \alpha_1) & \sigma(\alpha_n, \alpha_2) & \dots & \sigma(\alpha_n, \alpha_n) \end{bmatrix}$$

- Perform a singular value decomposition (SVD) on R:

$$R = USV^T$$

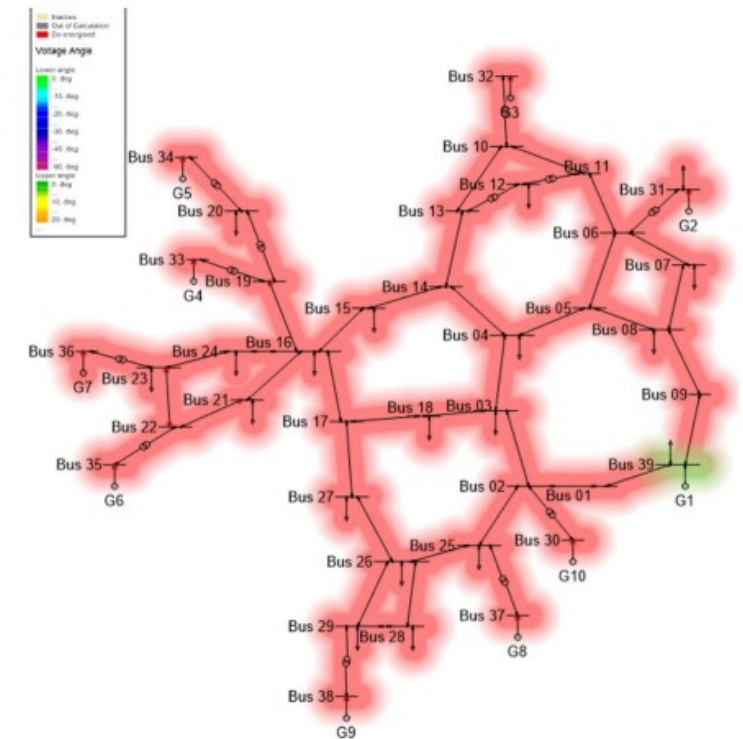
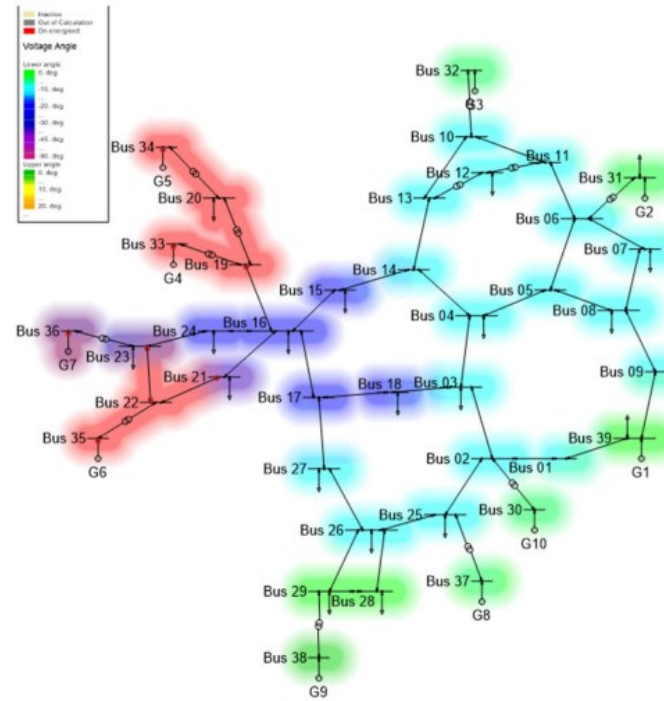
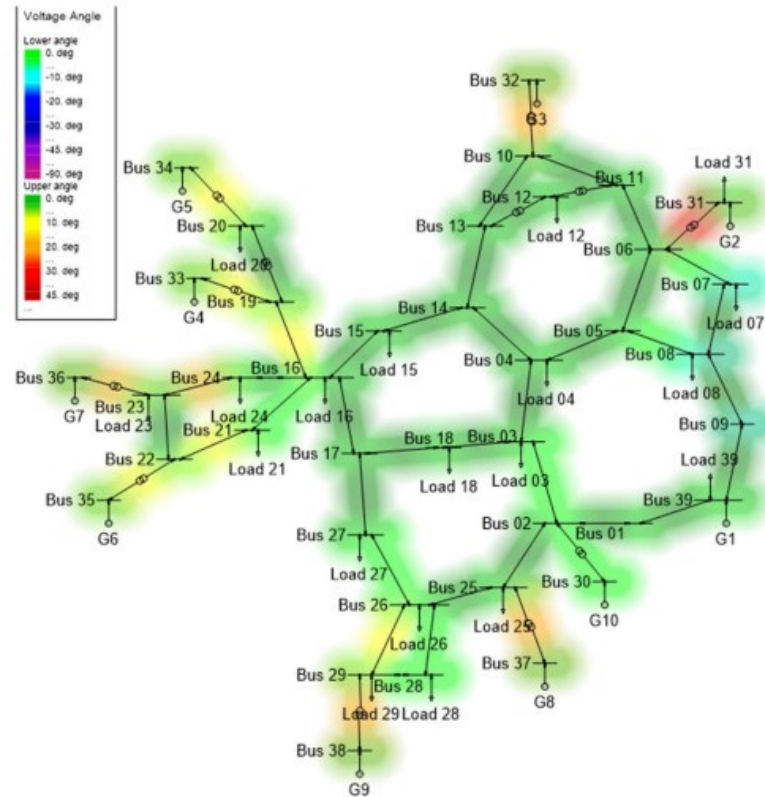
where U and V are distinct orthogonal matrices, and S is a sparse diagonal matrix with singular values of R

$$S = \begin{bmatrix} \sigma_i & 0 & \dots & 0 \\ 0 & \sigma_j & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n \end{bmatrix}$$



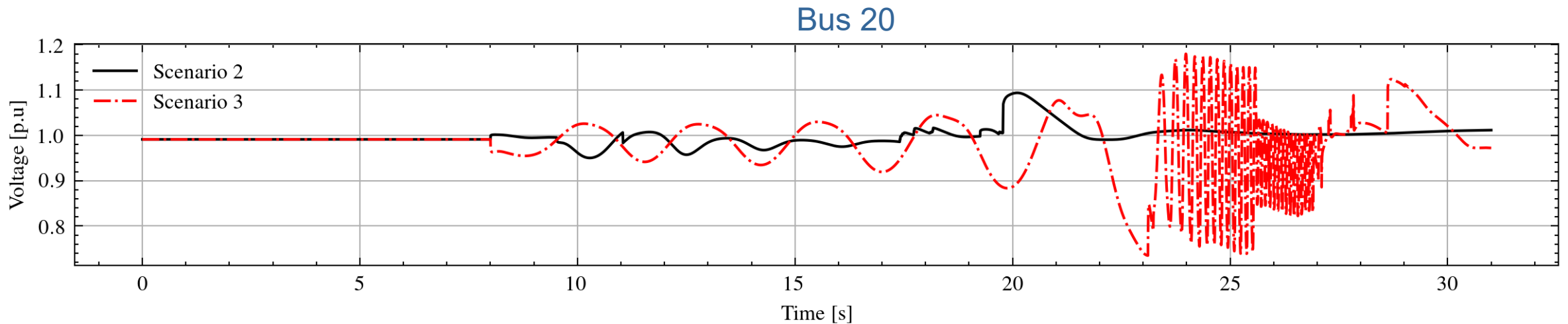
3. Experimental Results

Power System State Transition During Cascading Failures

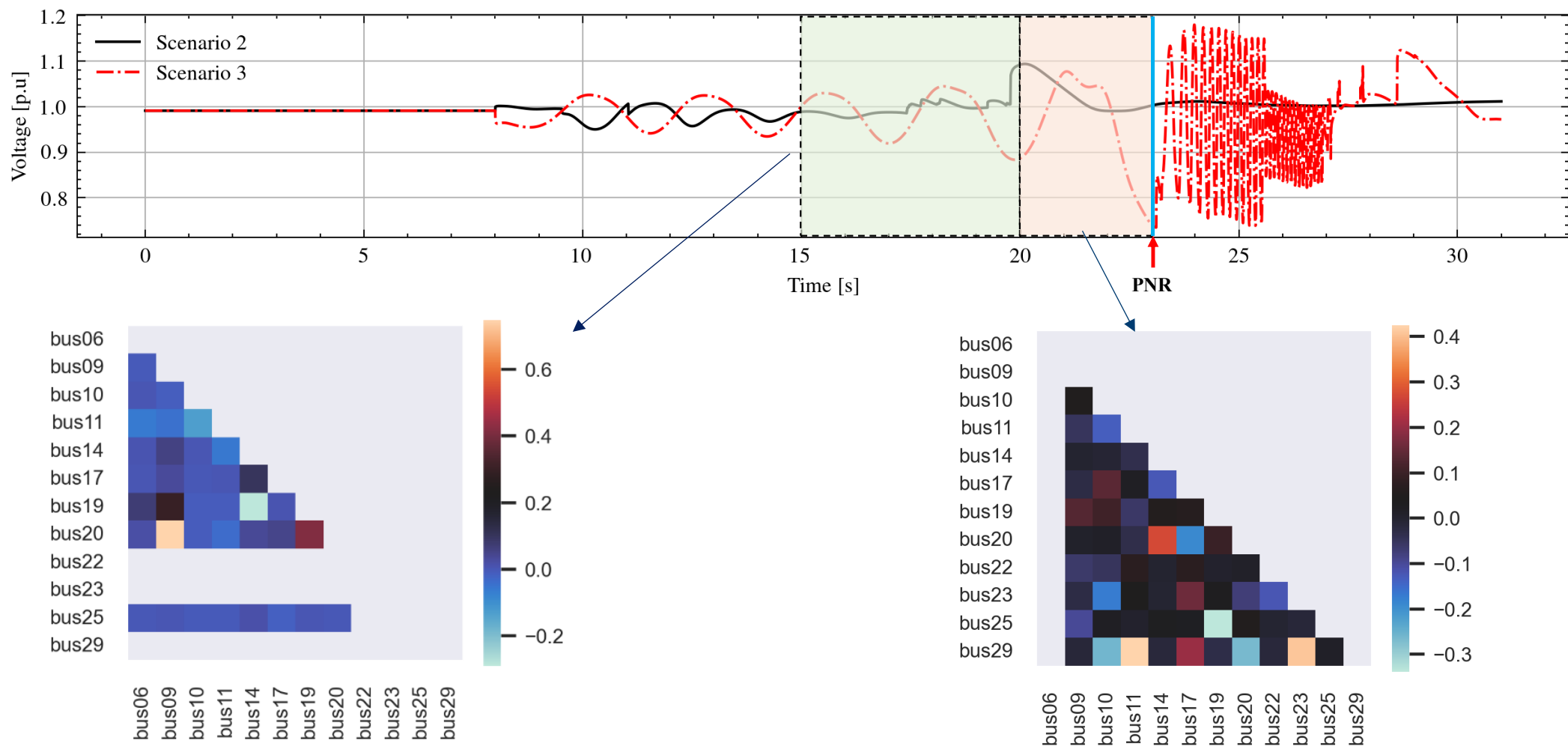


Point of No Return Identification

- Scenario 2: switching attack line 05-06 → instabilities
- Scenario 3: substation 2 attack → cascade → blackout

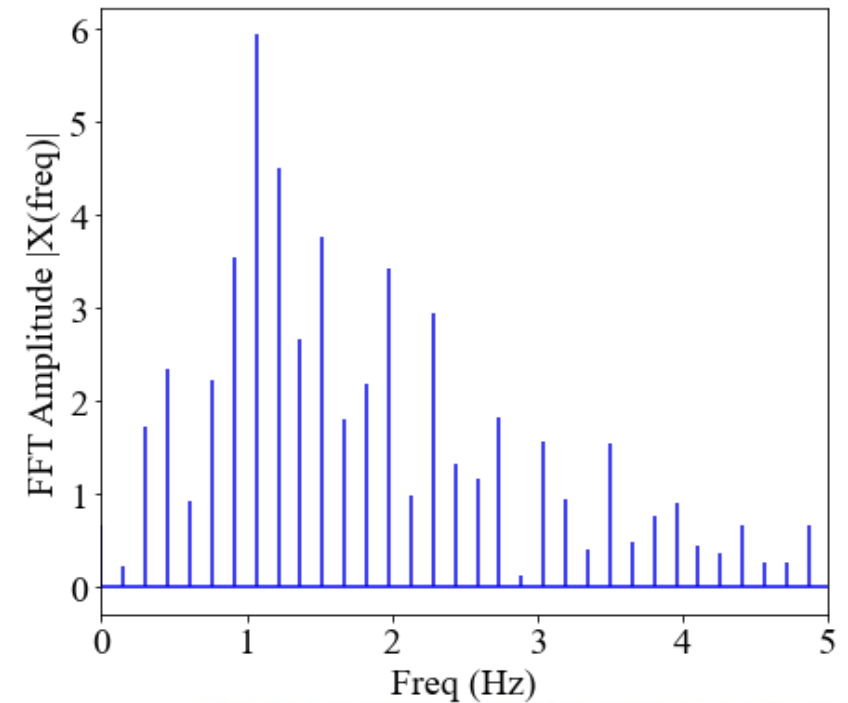
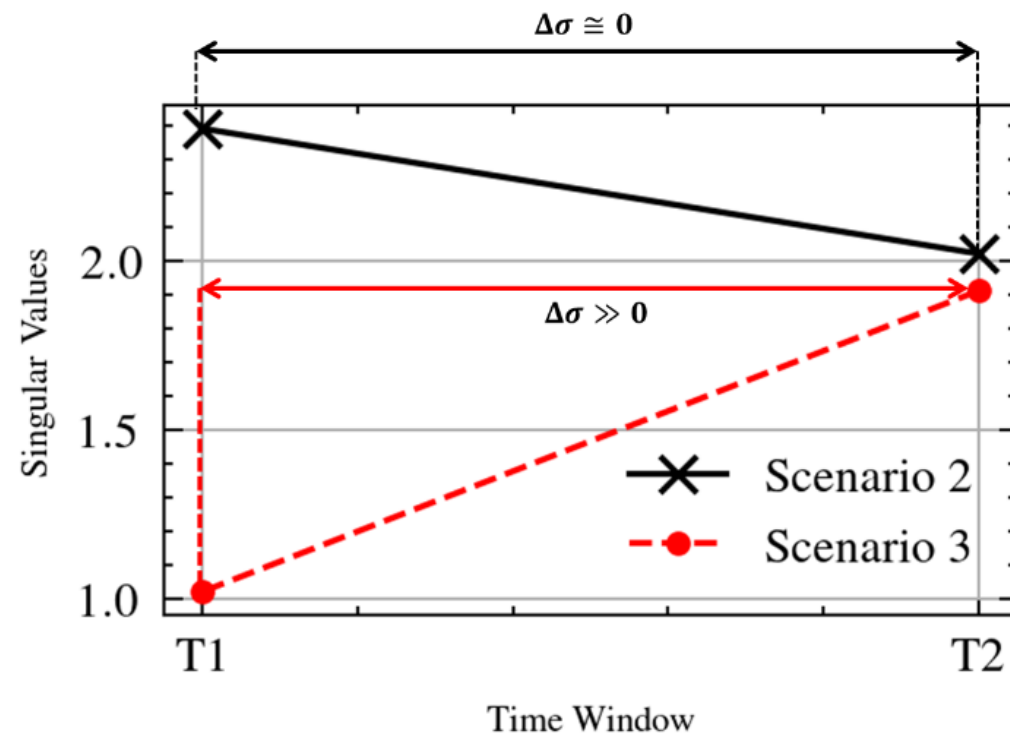


Point of No Return Identification

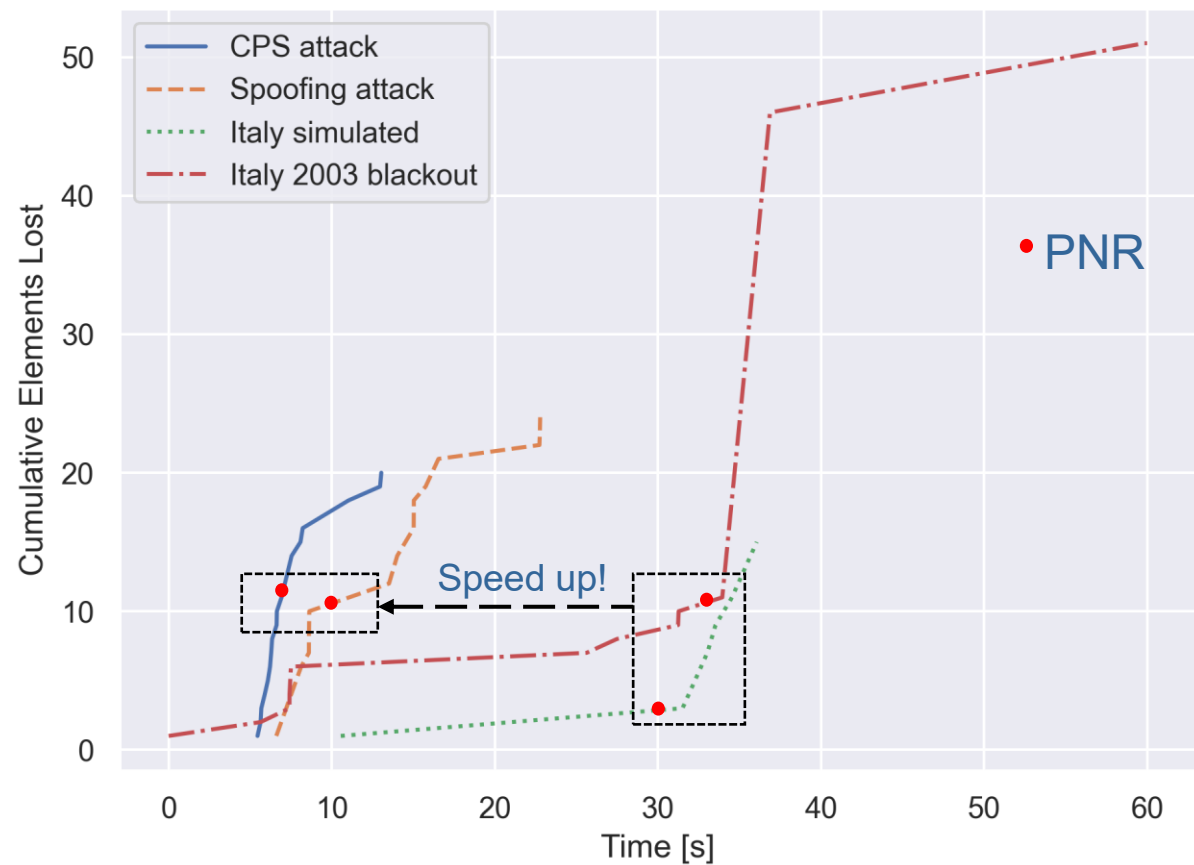


Point of No Return Analysis

- Approaching PNR, system moves between points of instability
- Cyber attacks can excite multiple frequency modes > 1 Hz



Acceleration Mechanism



Scenario	PNR (s)	γ	Load Lost (MW)
CPS attack	10	2.2	4000
Spoofing	6.5	8.6	6000
Italy blackout	31	0.1	2500

- Cyber attacks may induce **acceleration** of cascading failure mechanism ~ **3x**

Conclusions and Wrap-Up



Cause and Accelerate

Cyber attacks **cause and accelerate** cascading failures



Analysis

Time-frequency analysis:
promising solution to study
dynamical system behaviour



Mitigation

Cyber-physical defense
strategies needed

Thank You



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